

PROJECT SPECIFICATION: ENTERPRISE E-COMMERCE PLATFORM

Target Cloud Provider: Google Cloud Platform (GCP)

Target Region: us-central1

1. WORKLOAD & COMPUTE REQUIREMENTS:

- Containerized web API and frontend microservices running on Cloud Run.
- High availability with automatic autoscaling between 2 and 20 instances.
- Serverless VPC connector for private database and backend connectivity.

2. DATA STORAGE REQUIREMENTS:

- Relational PostgreSQL 15 database on Cloud SQL for order and user data.
- High availability (regional failover), private IP only, automated daily backups.
- Cloud Storage bucket for product images and static media assets.

3. NETWORKING REQUIREMENTS:

- Custom VPC network with dedicated private subnets for apps and databases.
- Cloud NAT and Cloud Router to allow instances outbound access without public IPs.
- External Application Load Balancer with SSL certificates.

4. SECURITY & COMPLIANCE REQUIREMENTS:

- CMEK (Customer-Managed Encryption Keys) using Cloud KMS for Cloud SQL and GCS.
- Cloud Armor security policy with WAF rules mitigating OWASP Top 10 vulnerabilities.
- Strict least-privilege IAM service accounts for Cloud Run and Cloud SQL access.

5. INFRASTRUCTURE AS CODE:

- Complete, modular Terraform (HCL) manifests deploying all resources.